Exercise 12-1.
請寫一個 Python 程式 `week12_1.py`
題目要求:
Given an array nums of size n, return the majority element.

The majority element is the element that appears more than ⌊n / 2⌋ times. You may assume that the majority element always exists in the array.

回傳格式請使用Tuple: **(majority_element,** 出現次數**)**

(範例輸入輸出參考下圖, 綠色表示系統輸出, 黑色表示使用者輸入)

```
None
Example 1:
Input: nums = [3,2,3]
Output: (3, 2)

Example 2:
Input: nums = [2,2,1,1,1,2,2]
Output: (2, 4)
```

```python
import ast
from collections import Counter

s = input("nums = ")
lst = ast.literal_eval(s) #可以辨識輸入的格式並轉換成不同格式 本題是轉換成list
count = Counter(lst)
print(Counter(lst).most_common()[0]) #輸出最多的單位、數量
```

```
C:\Users\benss\PycharmProjects\Pyt
nums = [3, 2, 3]
(3, 2)
```

```
C:\Users\benss\PycharmProjects\PythonProject\.
nums = [2, 2, 1, 1, 1, 2, 2]
(2, 4)

Process finished with exit code 0
```

有使用 AI

Exercise 12-2.
請寫一個 Python 程式 week12_2.py
題目要求:
Given two integer arrays nums1 and nums2, return an array of their intersection. Each element in the result must appear as many times as it shows in both arrays and you may return the result in any order.

None

```
Example 1:
Input: nums1 = [1,2,2,1], nums2 = [2,2]
Output: [2,2]

Example 2:
Input: nums1 = [4,9,5], nums2 = [9,4,9,8,4]
Output: [4,9]
Explanation: [9,4] is also accepted.
```

```python
import ast

s = input("nums1 = ")
p = input("nums2 = ")
fin = []
lst1 = ast.literal_eval(s)
lst2 = ast.literal_eval(p) #像上一個題目 轉換成list
set1 = set(lst1)
set2 = set(lst2) #分別轉換成集合
for i in set1:
    if i in set2: #如果在第一個集合裡面的元素也在第二個裡面
        n = min(lst1.count(i), lst2.count(i)) #取list中那個元素的數量 並取較小的那個
        fin += [i] * n #元素乘以共有的最小數量 加到list中

print(fin)
```

```
C:\Users\benss\PycharmP
nums1 = [1, 2, 2, 1]
nums2 = [2, 2]
[2, 2]
```

```
C:\Users\benss\PycharmProjects\Pythor
nums1 = [4, 9, 5]
nums2 = [9, 4, 9, 8, 4]
[9, 4]
```

除了沿用上一題轉換成 list 的功能 其餘未使用 AI